

Quantitative Investing

PROBLEM SET 3

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Contents. Part I tests the CAPM on the 49 value-weighted industry portfolios. Part III contains the three standalone AI exercises and the capstone, a cross-sectional test of U.S. sector ETFs. Part II repeats the tests on the 25 size and book-to-market portfolios and examines Roll's critique using in-sample and out-of-sample tangency portfolios.

PART I: 49 INDUSTRY PORTFOLIOS

Data and conventions. All figures are in percent per month. Observations coded -99.99 or -999 are treated as missing. Industry returns are total returns, while the market proxy is reported as $R_M - R_f$. We therefore subtract the monthly riskless rate from each industry return and work in excess returns throughout. Six industries have no data before the late 1960s, so all 49 series are populated only from July 1969. We report the balanced July 1969 – June 2026 panel ($T = 684$, $N = 49$) as our primary specification. The full July 1926 – June 2026 sample ($T = 1200$, N from 40 to 49) is reported as a robustness check. The sample mean of $R_M - R_f$ is 0.6116 on the balanced sample and 0.6952 on the full sample.

Question (a)

In the cross-sectional regression $R_i = \gamma_0 + \gamma_M \beta_{iM} + \eta_i$ with $\beta_{iM} = \text{cov}(R_i, R_M) / \sigma^2(R_M)$, each observation is a portfolio, so γ_0 is the expected return on a portfolio with zero market beta and γ_M is the expected return per unit of beta.

Sharpe/Lintner. With unrestricted borrowing and lending at the riskless rate, $E[R_i] = R_f + \beta_{iM}(E[R_M] - R_f)$, so

$$\gamma_0 = R_f, \quad \gamma_M = E[R_M] - R_f.$$

Both are exact point predictions. A portfolio with $\beta_{iM} = 0$ bears no systematic risk and must earn the riskless rate, which fixes the intercept. The market has $\beta_{iM} = 1$, so the line must pass through $(1, E[R_M])$, which fixes the slope at the market risk premium.

Black. Without riskless borrowing and lending, R_f is replaced by the expected return on the zero-beta portfolio R_z , the minimum-variance portfolio uncorrelated with the market, giving $E[R_i] = E[R_z] + \beta_{iM}(E[R_M] - E[R_z])$ and

$$\gamma_0 = E[R_z], \quad \gamma_M = E[R_M] - E[R_z].$$

Because $E[R_z]$ is not observable, this version predicts no specific intercept. It restricts only $\gamma_M > 0$ and $\gamma_0 < E[R_M]$, and is therefore weaker and harder to reject: Sharpe/Lintner is Black plus the restriction $E[R_z] = R_f$. Both imply $\gamma_0 + \gamma_M = E[R_M]$, so they differ only in where the security market line is anchored.

Because all tests use excess returns, the Sharpe/Lintner null becomes $\gamma_0 = 0$ with γ_M equal to the average market excess return, and Black predicts $\gamma_0 = E[R_z] - R_f$, unrestricted in sign but generally positive.

Question (b)

Each industry's beta is estimated once by OLS over the full sample and held fixed thereafter; the estimates range from 0.516 (Utilities) to 1.514 (Software), with mean 1.039. In each of the 684 months we then estimate $R_{it} = \gamma_{0t} + \gamma_{Mt}b_{iM} + \eta_{it}$ across the 49 industries. The monthly slopes are highly volatile, with a standard deviation of 7.13 against a mean of 0.035, and are positive in only 48.4% of months. Finally we average the monthly estimates, using

$$\hat{\gamma}_j = \frac{1}{T} \sum_{t=1}^T \hat{\gamma}_{jt}, \quad s(\hat{\gamma}_j) = \frac{\sigma(\hat{\gamma}_{jt})}{\sqrt{T}},$$

so that the standard error is the time-series standard deviation of the monthly estimates divided by $\sqrt{T}$, not the cross-sectional OLS standard error from any single month.

Table 1: Fama-MacBeth estimates, 49 industry portfolios, percent per month. The final column gives the value each coefficient should take if the CAPM holds, with $\gamma_0 = 0$ under Sharpe/Lintner in excess-return space.

	Estimate	Std. Error	t	p	CAPM
<i>Balanced, July 1969 – June 2026, $T = 684$</i>					
γ_0	0.6119	0.2191	2.79	0.005	0
γ_M	0.0352	0.2725	0.13	0.897	0.6116
<i>Full, July 1926 – June 2026, $T = 1200$</i>					
γ_0	0.5160	0.1714	3.01	0.003	0
γ_M	0.2490	0.2315	1.08	0.282	0.6952

Can we reject mean-variance efficiency? Yes. Both restrictions fail: the intercept is significantly positive where it should be zero, and the slope is indistinguishable from zero and significantly below the realized market premium ($t = -2.12$). Taken together these describe a single pattern, the flat security market line, which is too high at $\beta = 0$ and too shallow thereafter. That pattern licenses the stronger conclusion, because the exact linear relation between $E[R_i]$ and β_{iM} holds if and only if the proxy is mean-variance efficient with respect to the test assets. The positive intercept alone would be consistent with Black, which permits $E[R_z] > R_f$, but that reading requires the zero-beta rate to absorb essentially the entire market premium while beta goes unpriced. The full sample gives the same conclusion with slightly less extreme flattening.

AI integration: code generation. We first asked an LLM to explain three aspects of the procedure, limiting it to three lines each, so the omissions below reflect our prompt rather than its knowledge.

- **Fixed betas.** It answered that betas need time-series variation to estimate and that short rolling windows would let noise correlate mechanically with the same month's return. This matches our understanding. We would add that fixed betas also make the monthly $\hat{\gamma}_{Mt}$ comparable across months.
- **Standard errors.** It correctly identified cross-sectional correlation in residuals as the source of the difference and predicted correctly that the Fama-MacBeth errors would be larger. It did not note that the two methods give identical point estimates in a balanced panel.
- **Errors-in-variables.** It correctly described attenuation of the slope and the offsetting upward bias in the intercept, without adding that Fama-MacBeth does not fix this, since the same estimated betas enter every month.

We then asked it to implement the procedure and ran its script ourselves.

- (i) The delivered code was correct; the errors it reported were from earlier drafts, namely a 36-month rolling beta, a missing $\sqrt{T}$ divisor, and an assumed fixed cross-section of 49. Two corrections were ours: it ran the full rather than the balanced sample, and it described the missing codes as “−99.99 or −999” when only −99.99 occurs here.
- (ii) Yes. It divides the time-series standard deviation by $\sqrt{T}$, and our independent run reproduces its standard errors exactly. It contrasted this against a pooled panel OLS rather than against the cross-sectional regression the assignment describes.
- (iii) Yes on substance, but this assessment is the model’s own, so we treat it as a claim and verified it by re-running the script against our data.

Question (c)

Table 2: Fama-MacBeth compared with OLS on time-averaged returns, balanced sample. The point estimates are identical; only the standard errors differ.

	γ_0		γ_M	
	Estimate	Std. Error	Estimate	Std. Error
(b) Fama-MacBeth	0.6119	0.2191	0.0352	0.2725
(c) OLS on averages	0.6119	0.1154	0.0352	0.1086

Are the estimates different? No. Because the betas are fixed and the panel is balanced, every monthly regression uses the same regressor, so each monthly coefficient is a weighted average of that month’s returns with constant weights. Averaging those coefficients over time is the same calculation as regressing the time-averaged returns on beta. The estimates coincide exactly only in the balanced panel; in the full sample the number of industries varies across months, so the two methods weight months differently.

Are the standard errors different, and which method is superior? The Fama-MacBeth standard errors are roughly twice as large, because the two methods measure different sources of uncertainty. The cross-sectional regression treats the average returns as fixed and uses only the dispersion of the 49 residuals, which averaging over time has already made small. Fama-MacBeth instead uses the month-to-month variation in the estimated coefficients, which is large. Fama-MacBeth is superior: since the point estimates are the same, only the standard errors matter, and the cross-sectional regression assumes residuals are independent across industries. That assumption is not reasonable here, because industries share a common market shock. On the full sample this matters for the conclusion, as the cross-sectional regression makes $\hat{\gamma}_M$ appear significant while Fama-MacBeth does not.

Question (d)

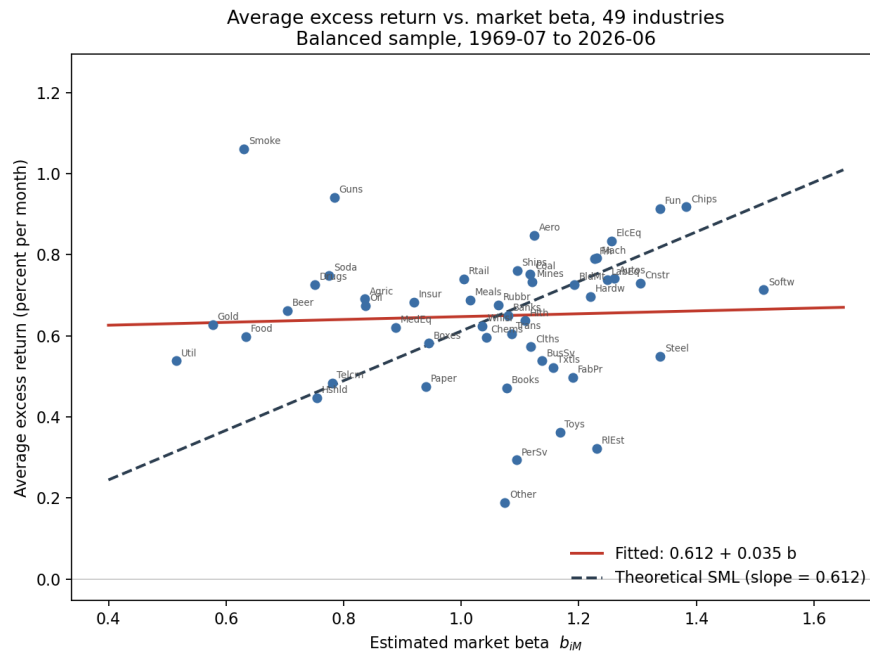


Figure 1: Average excess return against estimated beta, 49 industry portfolios, July 1969 – June 2026. Solid line: fitted cross-sectional regression, $\bar{R}_i = 0.612 + 0.035 b_{iM}$ with $R^2 = 0.002$. Dashed line: theoretical SML through the origin with slope 0.612.

Does the plot resemble a positive relationship? Barely. The fitted line is almost flat, and beta explains essentially none of the cross-sectional variation in average returns. Low-beta industries such as Utilities, Gold and Food earn about as much as high-beta industries such as Software, Chips and Steel. The vertical scatter at any given beta is much wider than the variation explained by beta itself, which is visible around $b_{iM} \approx 1.1$, where average returns range from Other near the bottom of the chart to Ships and Coal near the top.

What should the plot look like? Under the CAPM the points should lie along an upward-sloping straight line. In excess-return space that line passes through the origin, because a zero-beta portfolio should earn no premium, and its slope should equal the average market excess return. Deviations should be small and unsystematic, because beta is supposed to be the only characteristic that is priced. Instead the fitted line is far too high at $\beta = 0$ and far too shallow, so it crosses the theoretical line near $\beta = 1$ and diverges from it in both directions. This is the flat security market line of part (b) shown graphically.

AI integration: SML prediction. Asked what the scatter should look like and what the most common deviation is, the LLM answered that the points should lie on a line through the origin with slope equal to the market premium, that the estimated line is typically too flat with a positive intercept, and that this reflects leverage constraints under Black’s zero-beta CAPM, citing the Frazzini-Pedersen “betting against beta” result, and omitted risk factors such as size and value.

- (i) The empirical SML is much flatter than the theoretical one. The model predicted the direction correctly but understated the degree, describing a line that still slopes upward, whereas our slope is statistically indistinguishable from zero.

- (ii) It mentions all three, and its account of the leverage mechanism is correct. It does not mention errors-in-variables in the estimated betas, which produces the same flattening for purely statistical reasons, or Roll's critique, under which a flat line may indicate a poor market proxy rather than a failed model. The response also referred to our results, so it was not predicting blind.
- (iii) No. Beta explains almost none of the cross-sectional variation, and industries with very different betas earn similar average returns. Utilities and Software have the lowest and highest betas in the sample, yet their average returns differ by little.

Question (e)

In the extended regression

$$R_i = \gamma_0 + \gamma_M \beta_{iM} + \gamma_{size} \ln(\text{size}) + \gamma_{B/M} \ln(\text{BE/ME}) + \eta_i,$$

the CAPM predicts $\gamma_{size} = \gamma_{B/M} = 0$. Expected return depends only on covariance with the market, so beta is a sufficient statistic and no other characteristic should have explanatory power once beta is included. Size and book-to-market are firm characteristics rather than measures of systematic risk, so they should be priced only if they proxy for beta, and beta is already controlled for. The predictions for γ_0 and γ_M are unchanged from part (a).

This regression tests the Fama-French (1992) critique. A significantly negative γ_{size} would indicate a size premium and a significantly positive $\gamma_{B/M}$ a value premium. Either result rejects the CAPM, regardless of the estimate of γ_M .

Question (f)

Using the same fixed betas, we estimate each month

$$R_{it} = \gamma_{0t} + \gamma_{Mt} b_{iM} + \gamma_{size,t} \ln(\text{size}_{t-1}) + \gamma_{B/M,t} \ln(\text{BE/ME}_{t-1}) + \eta_{it},$$

with size lagged one month and book-to-market taken from the previous year and held constant within the year. Eleven negative book-equity observations are masked, since the logarithm is undefined for them.

Why must the characteristics be lagged? They must be known to investors before the return is realized. Size is measured by market capitalization, which moves with the return itself, so contemporaneous size would place part of the month's return on both sides of the regression and create a spurious relationship. Book-to-market has the same problem through its market value denominator, and annual accounting data is not public until several months after the fiscal year ends. Lagging makes the regression a test of whether information available at the start of the month predicts the return earned over it.

Table 3: Extended Fama-MacBeth estimates, balanced sample, July 1969 – June 2026 ($T = 684$), percent per month. The average cross-sectional R^2 rises from 0.09 in part (b) to 0.18.

	Estimate	Std. Error	t	CAPM
γ_0	0.4045	0.4988	0.81	0
γ_M	0.0328	0.2595	0.13	0.6116
γ_{size}	0.0428	0.0497	0.86	0
$\gamma_{B/M}$	0.1568	0.0805	1.95	0

Can we reject mean-variance efficiency? Yes, for the same reason as in part (b): the slope on beta is still indistinguishable from zero and significantly below the average market excess return ($t = -2.23$). Adding the characteristics does not restore a risk-return relationship.

The characteristics themselves give only weak evidence against the CAPM. The coefficient on book-to-market is positive and marginally significant, which is consistent with a value premium, while the coefficient on size is small and insignificant. This is not surprising for industry portfolios, which are not sorted on these characteristics and therefore show limited cross-sectional variation in them; the 25 portfolios in Part II provide a sharper test. The intercept is no longer significant once the characteristics are included, which suggests the zero-beta return found in part (b) is partly absorbed by book-to-market rather than being a separate effect. Over the full sample both characteristics are insignificant and the intercept is significantly positive, with the rejection again coming from the slope on beta ($t = -2.83$).

AI integration: interpreting size and B/M premia. The prompt specified in the assignment asserts significant premia on both characteristics, which our results do not show, so we ran it as written and then supplied our actual table, to test whether the model would confirm the premise it was given. Asked the specified question it identified the rejection at once, noting that “formally, you’re rejecting the joint restriction $\gamma_{size} = \gamma_{B/M} = 0$, which the CAPM requires,” and offered three interpretations: that the characteristics proxy for priced risks a single market index does not capture, following Fama and French; that “it’s the characteristics themselves, not the covariance structure, that predict returns,” following Daniel and Titman, which points to mispricing; and that a significant loading is “also consistent with the market proxy simply being wrong,” following Roll.

- (i) Yes. It identified the rejection immediately and stated the restriction the CAPM imposes.
- (ii) Yes, and more fully than required, contrasting the risk-based reading with the mispricing and invalid-proxy alternatives.
- (iii) Only partly. It noted that the characteristics are known before the return is earned, but did not make the point that lagging solves the econometric problem without settling whether the characteristics represent genuine risk factors.

Given our actual table it declined to claim a size effect and identified the beta slope as the main source of rejection, though it misstated $t = 1.95$ as roughly 5% one-sided and 10% two-sided when the correct values are 2.6% and 5.2%. Asking the model to interpret real output proved a better test than asking it to explain a result stated in the prompt.

PART II: 25 SIZE AND BE/ME PORTFOLIOS

Sample. 1969-07 to 2026-06, $T = 684$, $N = 25$. Estimates and standard errors: percentage points per month. Mean market excess return: 0.6116.

Question (1), repeating Part I(b)

Table 4: Extended Fama–MacBeth estimates, 25 size and BE/ME portfolios.

	Estimate	FMB SE	t -stat	p -value
gamma_0	1.3554	0.3998	3.390	0.0007
gamma_M	-0.5758	0.4290	-1.342	0.1800

The t -statistics test against zero. Average monthly R-squared is 0.2425 with beta alone and 0.5288 for the extended model. For the slope restriction $E(\gamma_{M,t} - \text{MktRF}_t) = 0$ the mean difference is -0.8761 with SE 0.2629, giving $t = -3.3320$ and $p = 0.0009$.

Using fixed full-period betas and 684 monthly cross-sectional regressions, we obtain $\bar{\gamma}_0 = 1.3554$ (FMB SE = 0.3998, $t = 3.390$) and $\bar{\gamma}_M = -0.5758$ (FMB SE = 0.4290, $t = -1.342$), in percent per month. Standard errors use $\text{sd}(\hat{\gamma}_{j,t})/\sqrt{T}$.

The significantly positive intercept rejects the zero-intercept restriction. The slope also falls below the average market excess return of 0.6116%; the difference has $t = -3.0316$ and $p = 0.0025$. These results reject the Sharpe–Lintner CAPM restrictions and provide evidence against the proxy being the tangency portfolio for these assets. The insignificant slope relative to zero is not, by itself, grounds for rejection.

Question (1), repeating Part I(c)

Table 5: Fama–MacBeth and single cross-sectional regression estimates compared. Cross-sectional R-squared: 0.1987. $N = 25$.

	FMB est.	FMB SE	FMB t -stat	OLS est.	OLS SE	OLS t -stat
gamma_0	1.3554	0.3998	3.390	1.3554	0.2619	5.1759
gamma_M	−0.5758	0.4290	−1.342	−0.5758	0.2411	−2.3884

The single cross-sectional regression gives the same coefficients as Fama–MacBeth: $\gamma_0 = 1.3554$ and $\gamma_M = -0.5758$, with $R^2 = 0.1987$ and $N = 25$. Equality follows because the panel is balanced and the regressors are fixed: averaging monthly OLS coefficients is equivalent to regressing average returns on the same betas.

The OLS standard errors, 0.2619 and 0.2411, are smaller than the FMB standard errors, 0.3998 and 0.4290. OLS measures dispersion around the average-return regression, whereas FMB measures variation in monthly coefficient estimates. FMB is preferable for inference on average premia here because it incorporates that time-series variation and accommodates contemporaneous correlation across portfolio returns. It does not eliminate error from estimated betas.

Question (1), repeating Part I(d)

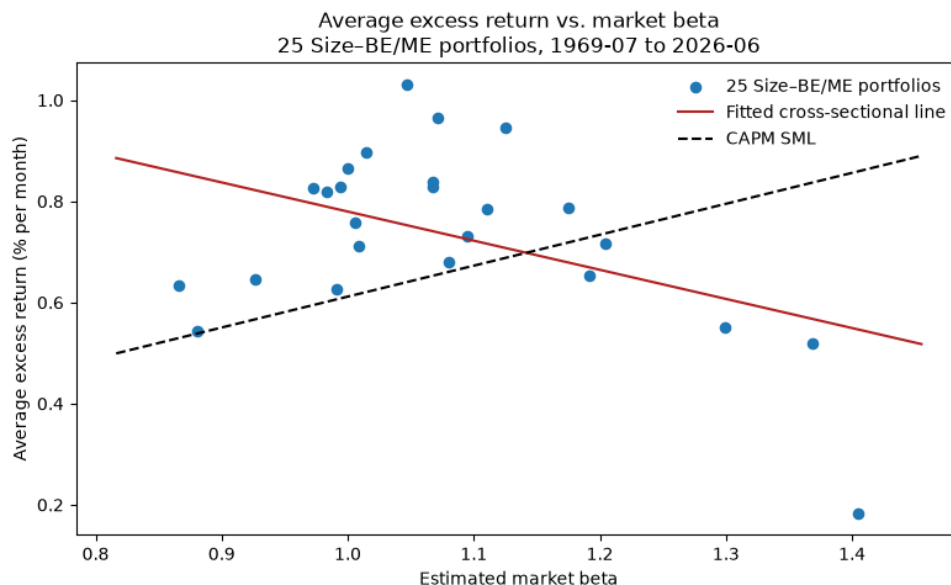


Figure 2: Average excess return against estimated market beta, 25 size–BE/ME portfolios. Fitted line: $1.3554 + (-0.5758) \times \text{beta}$. CAPM SML: $0.6116 \times \text{beta}$. Cross-sectional R-squared: 0.1987.

The plot shows a negative fitted relationship:

$$R_i^e = 1.3554 - 0.5758 \hat{\beta}_{iM}.$$

The CAPM instead predicts a line through the origin with positive slope 0.6116. The portfolios are widely scattered around the fitted line, with $R^2 = 0.1987$, so market beta explains little of their average-return variation. The plot is inconsistent with the predicted SML, although visual evidence alone is not a formal statistical test.

Question (1), repeating Part I(f)

Table 6: Extended Fama–MacBeth estimates (t -statistics test zero). Average monthly R-squared, beta only: 0.2425; extended: 0.5288. Slope restriction $E(\gamma_{M,t} - \text{MktRF}_t) = 0$: mean difference = -0.8761 , SE = 0.2629, $t = -3.3320$, $p = 0.0009$.

	Estimate	FMB SE	t -stat	p -value
gamma_0	1.1539	0.3392	3.4017	0.0007
gamma_M	-0.2646	0.3134	-0.8442	0.3989
gamma_size	-0.0194	0.0302	-0.6428	0.5206
gamma_BM	0.1573	0.0747	2.1055	0.0356

With lagged characteristics included, the estimates are $\gamma_0 = 1.1539$ ($t = 3.4017$), $\gamma_M = -0.2646$ ($t = -0.8442$), $\gamma_{size} = -0.0194$ ($t = -0.6428$), and $\gamma_{B/M} = 0.1573$ ($t = 2.1055$). Book-to-market has a positive premium significant at 5%, while size is insignificant. Average monthly cross-sectional R^2 rises from 0.2425 to 0.5288.

The positive book-to-market premium contradicts the CAPM prediction that characteristics have no additional explanatory power after controlling for beta. The intercept remains significantly positive, and the beta premium remains below the market premium ($t = -3.3320$, $p = 0.0009$). Thus, adding characteristics does not restore the Sharpe–Lintner CAPM restrictions.

Size is lagged one month and BE/ME uses the previous year’s value, held constant within each year. This follows the assignment’s information-timing convention and avoids using contemporaneous characteristics, though actual accounting-data availability would require publication dates.

Question (2)

2(a). Using the 684 monthly observations from July 1969 to June 2026, we calculate the sample mean vector $\hat{\mu}$ and covariance matrix $\hat{\Sigma}$ of the 25 portfolios’ excess returns. The tangency weights are

$$w_T = \frac{\hat{\Sigma}^{-1} \hat{\mu}}{\mathbf{1}' \hat{\Sigma}^{-1} \hat{\mu}}.$$

The weights sum to one, with short selling allowed.

Table 7: In-sample tangency weights and tangency betas.

Portfolio	Weight	Tang. beta	Portfolio	Weight	Tang. beta
Size1_BM1	−1.9858	0.0725	Size3_BM4	−0.2411	0.3253
Size1_BM2	0.6508	0.2841	Size3_BM5	0.2578	0.3834
Size1_BM3	−0.0034	0.2903	Size4_BM1	1.1460	0.2595
Size1_BM4	0.9189	0.3563	Size4_BM2	−0.6837	0.2694
Size1_BM5	0.9936	0.4092	Size4_BM3	−0.0040	0.3006
Size2_BM1	−0.2076	0.2061	Size4_BM4	0.4626	0.3274
Size2_BM2	0.8054	0.3119	Size4_BM5	−0.1819	0.3330
Size2_BM3	0.2286	0.3289	Size5_BM1	0.7176	0.2484
Size2_BM4	−0.0409	0.3435	Size5_BM2	0.0421	0.2559
Size2_BM5	−0.4743	0.3749	Size5_BM3	0.5150	0.2519
Size3_BM1	−0.6186	0.2183	Size5_BM4	−1.0900	0.2154
Size3_BM2	0.0531	0.3109	Size5_BM5	0.4014	0.3288
Size3_BM3	−0.6615	0.2819			

Weights are fractions of portfolio value and size and BE/ME groups increase from 1 to 5. Sum of weights 1.000000; gross exposure 13.3859; mean excess return 2.5204% per month; monthly volatility 6.0461%; monthly Sharpe ratio 0.4169; weighted average beta 1.000000.

Gross exposure,

$$\sum_i |w_i|,$$

is 13.3859, indicating substantial offsetting long and short positions.

2(b) Returns, betas, and cross-sectional results. We construct the tangency portfolio's monthly excess return using the same fixed weights throughout:

$$R_{T,t}^e = \sum_{i=1}^{25} w_{T,i} R_{i,t}^e.$$

Its average excess return is 2.5204% per month, monthly volatility is 6.0461%, and the monthly Sharpe ratio is 0.4169.

We estimate each portfolio's beta by regressing its excess returns on $R_{T,t}^e$, including an intercept. Regressing average excess returns on these new betas gives

$$R_i^e \approx 0 + 2.5204 \hat{\beta}_{i,T}, \quad R^2 = 1, \quad N = 25.$$

The estimated intercept is effectively zero, while the slope equals the tangency portfolio's average excess return. All 25 portfolios lie essentially on the theoretical SML, so the fitted and theoretical lines overlap. The OLS standard errors are effectively zero; the resulting t -statistics are not economically meaningful because the fit is mechanically exact.

Table 8: Cross-sectional regression using the in-sample tangency portfolio as the market proxy.

	Estimate	OLS SE	t -stat
gamma_0	4.83988×10^{-15}	4.34718×10^{-15}	1.11334
gamma_M	2.52036	1.45191×10^{-14}	1.7359×10^{14}

Cross-sectional R-squared 1.000000, $N = 25$. Mean tangency excess return 2.5204% per month and maximum absolute residual 9.659×10^{-15} . Fitted line $0.0000 + (2.5204) \times \text{beta}$; CAPM SML $2.5204 \times \text{beta}$.

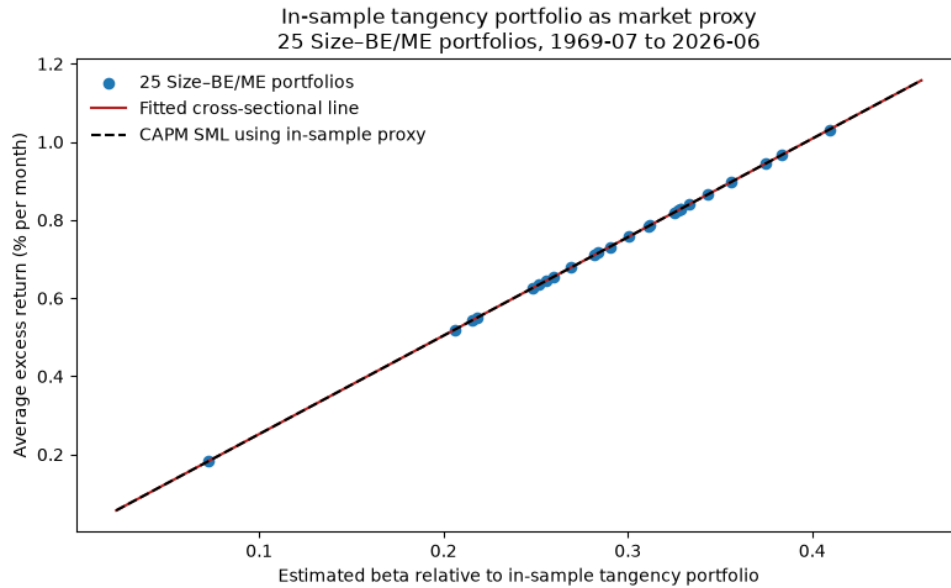


Figure 3: Average excess return against beta estimated against the in-sample tangency portfolio. Fitted line: $0.0000 + (2.5204) \times \text{beta}$. CAPM SML: $2.5204 \times \text{beta}$.

2(c). This perfect fit does not independently support the CAPM. The tangency portfolio is constructed to be mean–variance efficient using the same assets and observations used in the regression. Its weights satisfy

$$w_T \propto \hat{\Sigma}^{-1} \hat{\mu},$$

which implies

$$\hat{\mu}_i = \hat{\beta}_{i,T} \hat{\mu}_T.$$

Thus, the zero intercept, matching slope, and perfect fit follow from portfolio mathematics.

This illustrates Roll’s critique: CAPM tests depend on whether the chosen market proxy is mean–variance efficient. The true market portfolio includes all wealth and is not directly observable. Rejecting the CAPM with a particular proxy could therefore reflect a poor proxy, while obtaining a perfect fit with a deliberately constructed tangency portfolio does not establish that the true market portfolio is efficient.

Question (3)

3(a). Sample 1 contains odd-numbered months in even years and even-numbered months in odd years, giving 342 observations. Using only these observations, we estimate the mean vector and covariance matrix of excess returns and calculate

$$w_1 = \frac{\hat{\Sigma}_1^{-1} \hat{\mu}_1}{\mathbf{1}' \hat{\Sigma}_1^{-1} \hat{\mu}_1}.$$

The weights sum to 1, with gross exposure $\sum_i |w_{1,i}| = 15.4800$ and individual weights ranging from -1.6671 to 1.4736 . We hold these weights fixed and apply them to the 342 months in Sample 2. These returns are out of sample because none of the evaluation observations were used to estimate the weights.

3(b). Sample 2 contains even-numbered months in even years and odd-numbered months in odd years, also giving 342 observations. We estimate a separate tangency portfolio using only Sample 2's mean vector and covariance matrix, with the same normalization.

These weights sum to 1, have gross exposure of 17.2082, and range from -2.4853 to 1.3680 . We then apply them, unchanged, to Sample 1 returns. This reverses the estimation and evaluation samples, ensuring that neither half is evaluated using weights estimated from itself. The checks confirm that the samples do not overlap and together cover all 684 months.

Table 9: Out-of-sample tangency weights. Sample 1: 342 months; Sample 2: 342 months; total 684 months, no overlap and complete coverage.

Estimated from	Applied to	Weight sum	Gross exposure	Min weight	Max weight
Sample 1	Sample 2	1.0	15.4800	-1.6671	1.4736
Sample 2	Sample 1	1.0	17.2082	-2.4853	1.3680

3(c). We combine Sample 1 returns generated with w_2 and Sample 2 returns generated with w_1 , then sort them chronologically. The resulting OOS tangency series covers July 1969–June 2026 and has a mean excess return of 1.9754% per month, monthly volatility of 7.0300%, and monthly Sharpe ratio of 0.2810.

We use this combined excess-return series as the market proxy and estimate each portfolio's beta through a full-period time-series regression with an intercept. The cross-sectional regression of average excess returns on these betas is

$$R_i^e = 0.0684 + 3.2851 \hat{\beta}_{i,\text{OOS}} + \hat{\eta}_i.$$

Table 10: Cross-sectional regression using the out-of-sample tangency portfolio as the market proxy.

Coefficient	Estimate	OLS standard error	t -statistic
γ_0	0.0684	0.0228	2.9930
γ_M	3.2851	0.1091	30.1058

Returns are in percent per month. The regression has $R^2 = 0.9753$ and $N = 25$, and the mean out-of-sample market excess return is 1.9754% per month.

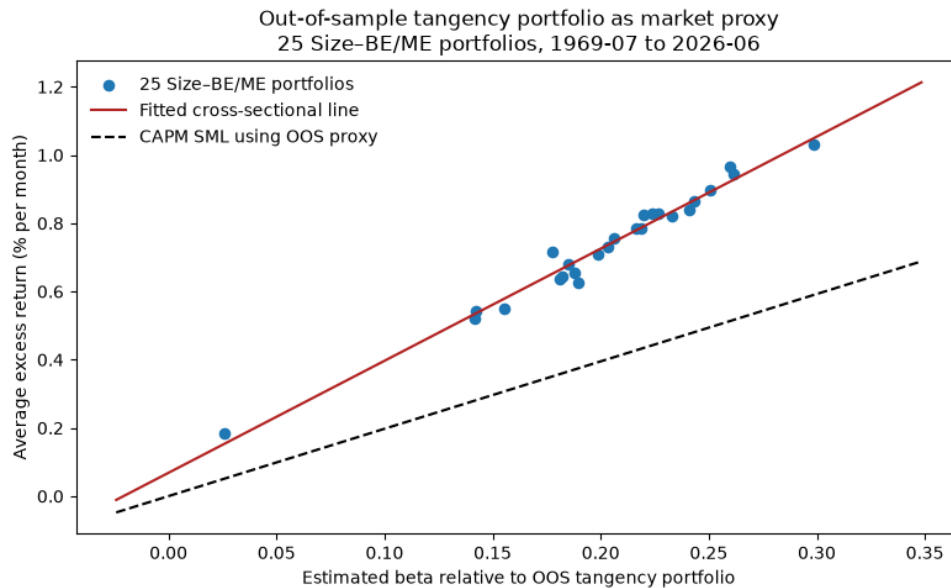


Figure 4: Average excess return against beta estimated against the out-of-sample tangency portfolio. Fitted line: $0.0684 + (3.2851) \times \text{beta}$. CAPM SML: $1.9754 \times \text{beta}$.

The plot shows a strong positive relationship: the portfolios cluster closely around the fitted line, which explains 97.53% of the cross-sectional variation in average excess returns. However, the fitted line lies above the theoretical SML, whose intercept is zero and slope is 1.9754. Thus, a close fit to the estimated line does not establish that the CAPM restrictions hold. The reported slope t -statistic tests equality to zero, not equality to 1.9754.

Each month's return is excluded from the sample used to estimate its portfolio weights. Because the two samples are interleaved in time, this is a held-out-sample exercise rather than a strictly forward-looking trading backtest.

Question (4)

LLM explanation obtained before comparing Q2 and Q3 Using an in-sample tangency portfolio as the market proxy makes the CAPM test almost tautological.

Suppose the same N assets are used both to construct the proxy and to test the CAPM. The tangency portfolio is chosen as

$$w_T \propto \Sigma^{-1}(\mu - r_f \mathbf{1}),$$

where μ and Σ are the sample mean returns and covariance matrix of those assets. By construction, this portfolio maximizes the sample Sharpe ratio and therefore lies on the sample mean-variance frontier.

The first-order condition implies

$$\mu - r_f \mathbf{1} \propto \Sigma w_T.$$

But Σw_T is exactly the vector of covariances between each asset and the tangency portfolio:

$$\text{Cov}(R_i, R_T).$$

Therefore,

$$E[R_i] - r_f = \beta_{iT}(E[R_T] - r_f), \quad \text{with} \quad \beta_{iT} = \frac{\text{Cov}(R_i, R_T)}{\text{Var}(R_T)}.$$

So the CAPM pricing relation holds for the test assets essentially by construction. The empirical procedure has selected the portfolio specifically to be mean-variance efficient relative to exactly those assets and exactly those sample moments. Finding small pricing errors or a strong beta-return relation therefore provides little independent evidence for the CAPM.

Connection to Roll (1977) This illustrates the core issue behind the Roll critique: empirically testing the CAPM is fundamentally a test of whether the chosen market proxy is mean-variance efficient. The true CAPM market portfolio contains all relevant wealth and is unobservable, so testing the CAPM with a proxy is always partly a test of that proxy.

With an in-sample tangency proxy, the problem becomes particularly severe because efficiency has been imposed mechanically, rather than discovered empirically.

Why out-of-sample is more informative An out-of-sample tangency portfolio separates construction from evaluation. For example:

1. Estimate $\hat{\mu}$, $\hat{\Sigma}$, and tangency weights using Sample 1.
2. Fix those weights.
3. Evaluate the portfolio and the CAPM relation using Sample 2.

In Sample 2, the portfolio is not guaranteed to remain mean-variance efficient, because realized means and covariances need not equal the Sample 1 estimates. Thus the CAPM pricing relation no longer follows mechanically from the portfolio-construction algebra.

So the key distinction is:

- In-sample: efficiency is imposed by construction $\rightarrow$ CAPM can trivially appear to work.
- Out-of-sample: efficiency must survive in new data $\rightarrow$ the test contains genuine empirical information.

Table 11: In-sample and out-of-sample tangency proxies compared.

Statistic	Q2: In-sample	Q3: Out-of-sample
γ_0 (SE, t)	4.84×10^{-15} (4.35×10^{-15} , $t = 1.11$)	0.0684 (0.0228, $t = 2.99$)
γ_M (SE, t)	2.5204 (1.45×10^{-14} , $t = 1.74 \times 10^{14}$)	3.2851 (0.1091, $t = 30.11$)
Proxy mean excess return (CAPM slope)	2.5204	1.9754
$\gamma_M \div$ proxy premium	1.00	1.66
Cross-sectional R^2	1.000000	0.9753
Max absolute residual	9.66×10^{-15}	not reported
Proxy Sharpe ratio (monthly)	0.4169	0.2810

1. Comparing the results.

- γ_0 : In Q2 the intercept is zero to machine precision. The $t = 1.11$ is rounding noise, not a meaningful test. In Q3 the intercept is 0.0684% per month, about 0.82% per year, with $t = 2.99$, indicating a statistically significant common pricing error across the 25 portfolios.
- γ_M : In Q2 the slope equals the proxy's premium exactly, 2.5204. In Q3 the slope is 3.2851, about 1.66 times the proxy premium of 1.9754. The raw slopes should not be compared directly across Q2 and Q3 because each set of betas is measured against a different proxy.

What matters for the CAPM is whether the slope equals that proxy's own premium. It does in Q2 but not in Q3.

- Standard errors: Q2's standard errors are essentially zero because the regression residuals are essentially zero, so the enormous slope t -statistic is not economically informative. Q3's t -statistics use ordinary cross-sectional OLS standard errors, which do not account for beta-estimation error or cross-portfolio dependence. Also, $t = 30.11$ tests $\gamma_M = 0$, not the CAPM restriction $\gamma_M = 1.9754$.
- SML shape: In Q2 the fitted line, $R_i^e = 0.0000 + 2.5204\beta_i$, and the theoretical SML are the same line. In Q3 the fitted line, $R_i^e = 0.0684 + 3.2851\beta_i$, has both a higher intercept and a steeper slope than the theoretical SML, $R_i^e = 1.9754\beta_i$. Thus, for positive betas, the fitted Q3 line lies above the theoretical SML.
- Which is cleaner? Q2 fits the CAPM restrictions mechanically: the intercept is zero, the slope equals the proxy premium, and $R^2 = 1$. Both Q2 and Q3 fit much better than the value-weighted market proxy in Part I, where $\gamma_0 = 1.3554$, $\gamma_M = -0.5758$, and $R^2 = 0.1987$.

2. Why Q2 and Q3 differ, and the link to Roll (1977). The in-sample proxy is efficient by construction. Using excess returns, the tangency weights are

$$w_T = \frac{\hat{\Sigma}^{-1}\hat{\mu}}{\mathbf{1}'\hat{\Sigma}^{-1}\hat{\mu}}.$$

Let

$$c = \mathbf{1}'\hat{\Sigma}^{-1}\hat{\mu}.$$

Then the result follows in four steps:

1. Rearranging the tangency-weight formula gives $\hat{\Sigma}w_T = \hat{\mu}/c$.
2. $\hat{\Sigma}w_T$ is the vector of sample covariances between each portfolio and the tangency portfolio: $\widehat{\text{Cov}}(R_i, R_T)$.
3. The tangency portfolio's variance is $w_T'\hat{\Sigma}w_T = \hat{\mu}_T/c$.
4. Therefore,

$$\hat{\beta}_i = \frac{\widehat{\text{Cov}}(R_i, R_T)}{\widehat{\text{Var}}(R_T)} = \frac{\hat{\mu}_i}{\hat{\mu}_T}, \quad \text{which implies} \quad \hat{\mu}_i = \hat{\beta}_i \hat{\mu}_T.$$

This identity holds for every portfolio.

Q2 estimates the weights using the same 684 months and the same 25 portfolios that are subsequently used in the CAPM test. Because the OLS betas are also based on the same sample covariances and variances, the beta-pricing identity must hold essentially exactly. The maximum residual of 9.66×10^{-15} confirms this.

Why the CAPM can look successful even when the market is mismeasured. Roll's critique is that an empirical CAPM test is fundamentally a test of whether the chosen market proxy is mean-variance efficient. The true CAPM market portfolio contains all relevant wealth and cannot be directly observed.

The results illustrate the problem. With the observable value-weighted market proxy in Part I, the CAPM performs poorly. With a proxy constructed directly from the test assets in Q2, the fit becomes perfect. Nothing about the true market changed; only the proxy changed.

Therefore, Q2's apparent success mainly shows that the proxy was constructed to be mean-variance efficient relative to these 25 portfolios in this sample. A perfect fit does not independently

establish that the true market portfolio is efficient, while rejection using another proxy could reflect proxy error rather than failure of the CAPM itself.

Why the out-of-sample version is more informative. In Q3, each month's proxy return uses tangency weights estimated from the opposite half of the data. Therefore, the means and covariances used to construct the proxy are no longer the same moments used to evaluate it. The exact in-sample identity above no longer has to hold.

Estimation error therefore becomes important:

- Inverting the 25×25 covariance matrix in $\hat{\Sigma}^{-1}\hat{\mu}$ can magnify estimation noise.
- This produces large long and short positions. Gross exposure is 15.48 and 17.21 in the two estimation samples, with individual weights ranging roughly from -2.4853 to 1.4736 .
- Weights optimized using one half of the data may partly fit noise specific to that half. When applied to the other half, performance deteriorates. The proxy mean falls from 2.5204% to 1.9754% per month, volatility rises from 6.0461% to 7.0300%, and the monthly Sharpe ratio falls from 0.4169 to 0.2810.
- Because the OOS proxy is no longer guaranteed to lie on the evaluation-sample mean-variance frontier, pricing errors appear: $\gamma_0 > 0$, the fitted slope differs from the proxy premium, and $R^2 < 1$.

The remaining high R^2 of 0.9753 indicates that the beta-return relation remains strong out of sample, even though the exact in-sample CAPM identity disappears.

Limits of the Q3 test:

- The two samples alternate months and therefore span the same broad economic periods, so this is a relatively mild out-of-sample test.
- The second-pass average returns and betas are calculated using the combined sample, so the design is not equivalent to a completely separate future holdout period.
- The OOS proxy is still constructed from the same 25 test portfolios rather than from the true aggregate market portfolio. Therefore, Q3 makes the test more informative but does not eliminate Roll's critique.

3. Evaluating the LLM explanation. 1. Did it predict a steeper SML and lower pricing errors? Partly.

The LLM correctly predicted that an in-sample tangency proxy would generate small pricing errors and a very strong beta-return relationship. It actually understated the strength of the result: it called the exercise “almost tautological,” while Q2 produces an essentially exact fit with $R^2 = 1.000000$.

However, it did not explicitly predict a steeper SML. In fact, “steeper” is not the key criterion. Q3's fitted slope, 3.2851, is numerically steeper than Q2's 2.5204. The important CAPM restriction is instead whether the fitted slope equals the proxy's own mean excess return. That equality holds exactly in Q2 but fails in Q3.

2. Did it explain why the portfolio is efficient by construction? Yes. This is the strongest part of the LLM explanation.

It correctly argues that the tangency first-order condition implies

$$\mu - r_f \mathbf{1} \propto \Sigma w_T,$$

and recognizes that Σw_T is the vector of covariances between each test asset and the tangency portfolio. This leads directly to beta pricing.

There are two minor precision issues:

- It skips the intermediate step needed to determine the proportionality constant. Premultiplying by w_T' gives the connection between the constant, the tangency portfolio premium, and its variance.
- It writes the result using population expectations such as $E[R_i]$, whereas the exact identity here is an *in-sample* identity involving sample moments, such as $\hat{\mu}_i$. This distinction explains why the Q2 fit is exact in the observed sample but need not hold out of sample.

3. Did it anticipate why the out-of-sample results could deteriorate? In general terms, yes. The LLM correctly explained that moments in the evaluation sample need not equal the moments used to estimate the weights, so mean-variance efficiency is no longer guaranteed.

However, its explanation was less complete about the empirical pattern observed in Q3. It did not emphasize:

- estimation error amplified by $\hat{\Sigma}^{-1}$;
- the resulting extreme leveraged weights;
- the lower out-of-sample Sharpe ratio;
- the positive intercept and slope exceeding the proxy premium;
- the fact that R^2 nevertheless remains high at 0.9753;
- the exact two-way sample-splitting procedure used in Q3;
- or the fact that the OOS proxy is still not the true market portfolio, so Roll's critique remains relevant.

Overall, the LLM's explanation is conceptually correct and explains the exact in-sample fit in Q2 well. It is less complete in explaining the magnitude and specific pattern of the deterioration observed in Q3.

AI EXERCISE 1 — STRESS-TESTING LLM KNOWLEDGE OF FAMA-MACBETH ECONOMETRICS

LLM used: ChatGPT. The three questions were asked one at a time in a single conversation. Full responses: [shared transcript](#).

Question 1: Shanken (1992) Correction

Prompt. What is the Shanken (1992) correction and when should it be applied to Fama-MacBeth standard errors?

Summary of the LLM's Answer. The LLM explained that the second-pass Fama-MacBeth regression uses estimated betas, but conventional Fama-MacBeth standard errors treat them as known, so they overstate the precision of the estimated risk premia. It gave Shanken's adjusted covariance matrix, $V_{Shanken} = (1 + c)\Omega + \Sigma_f^*$ with $c = \lambda' \Sigma_f^{-1} \lambda$, compared with the conventional $V_{FM} = \Omega + \Sigma_f^*$, so corrected standard errors are larger and t -statistics smaller. It said the correction should be applied when the second-pass regressors are estimated betas, but generally not when they are observed characteristics such as size or book-to-market, and that it matters more when betas are noisy or the time series used to estimate them is short. It also noted that Shanken's result assumes conditional homoskedasticity and that the correction is different from Newey-West, which addresses serial correlation and heteroskedasticity.

Evaluation. The answer is mostly correct. The LLM correctly identifies the source of the problem (estimated betas treated as known), the direction of the error in conventional standard errors (too small), when the correction applies and when it does not, and the difference from Newey-West. Its general formula is consistent with the assignment's answer: with a single market factor, $c = \lambda^2/\sigma_M^2$, so the correction inflates the conventional variance by $(1 + \lambda^2/\sigma_M^2)$. However, it never reduces the formula to this one-factor case, so the key result is left implicit. Its statement that the correction matters more when the beta-estimation sample is short is imprecise: the size of the correction relative to the conventional standard error depends on c , the squared Sharpe ratio of the factor, not on the sample length. It also does not mention that c is small for a monthly market factor, so the correction is usually modest in practice.

My Correction. In the one-factor case, $c = \lambda^2/\sigma_M^2$, so the Shanken correction inflates the conventional Fama-MacBeth variance (apart from the factor-variance term) by $1 + \lambda^2/\sigma_M^2$, and because this term is the squared Sharpe ratio of the market rather than a function of sample length, it is usually small for monthly market data.

Question 2: Errors-in-Variables

Prompt. Does the Fama-MacBeth procedure fully solve the errors-in-variables problem caused by using estimated betas as regressors? If not, what bias remains?

Summary of the LLM's Answer. The LLM said Fama-MacBeth does not fully solve the errors-in-variables problem: the same noisy estimated betas are the regressors in every monthly cross-sectional regression, so averaging the monthly risk-premium estimates does not remove the measurement error. With one factor and classical measurement error, the estimated premium is attenuated toward zero by the factor $\text{Var}(\beta)/[\text{Var}(\beta) + \text{Var}(e)]$; with several factors, the estimates are jointly distorted and not every coefficient is necessarily biased toward zero. It added that the Shanken correction adjusts the standard errors but does not remove the bias in the coefficients, and that the bias shrinks as the first-stage beta estimates become more precise.

Evaluation. The answer is correct but incomplete. The LLM correctly explains why averaging does not remove the measurement error, identifies attenuation bias toward zero in the one-factor case, gives the attenuation factor, notes that the multi-factor case cannot always be signed, and usefully distinguishes the bias (which Shanken does not fix) from the standard errors (which it does). However, it does not state that the bias persists even in large samples, meaning that adding more assets does not reduce it. It also does not mention that attenuation of the slope biases the intercept upward, producing a flatter estimated security market line, or that grouping stocks into portfolios is the standard way to reduce beta measurement error.

My Correction. Because the same noisy betas enter every monthly regression, the estimated market premium γ_M is attenuated toward zero, and with a positive premium the intercept is biased upward, even when the number of assets is large, so the bias can only be reduced by estimating betas more precisely, for example by using portfolios.

Question 3: Serial Correlation in Monthly Gamma Estimates

Prompt. In a FMB regression where the same betas are used every month, are the monthly gamma estimates serially uncorrelated? Does this matter for the standard error calculation?

Summary of the LLM’s Answer. The LLM said that using the same betas every month does not by itself make the monthly gamma estimates serially correlated. Because the betas are fixed, each month’s estimate is $\hat{\gamma}_t = (\hat{B}'\hat{B})^{-1}\hat{B}'R_t$, so the gammas are serially correlated only if returns or regression disturbances are correlated over time, which can happen in real data. The usual standard error $s(\hat{\gamma}_t)/\sqrt{T}$ assumes no serial correlation; if the gammas are positively autocorrelated, it is usually too small and the t -statistics too large, and Newey-West/HAC standard errors should be used. It again distinguished Newey-West (serial correlation) from Shanken (estimated betas).

Evaluation. The answer is mostly correct. It identifies the mechanism behind the assignment’s answer: because the betas are fixed, each $\hat{\gamma}_t$ is a fixed linear combination of that month’s returns, so the gammas inherit whatever autocorrelation the returns have. It also correctly explains why this matters, the direction of the resulting error in the standard errors, and the Newey-West remedy. Its framing (fixed betas do not by themselves cause serial correlation) differs in emphasis from the assignment’s (fixed betas are why the gammas inherit return autocorrelation), but the two are consistent. However, it does not address magnitude: it presents Newey-West as the remedy without noting that autocorrelation in monthly returns, and therefore in the gammas, is typically mild.

My Correction. Because the betas are fixed, each monthly gamma is a fixed-weight portfolio return, so the gammas are serially correlated only to the extent that returns are, which is typically mild for monthly data, so the usual $s(\hat{\gamma}_t)/\sqrt{T}$ standard error is usually adequate, with Newey-West as a robustness check.

AI EXERCISE 2 — PREDICTING CROSS-SECTIONAL DEVIATIONS FROM THE SML

We asked an LLM which of the 49 industries it expected to fall above, below and near the security market line, and recorded its answer before comparing it with our results from Question (d).

(a) The LLM’s predictions.

- *Above the line:* Tobacco, Utilities, Beer & Liquor / Food.
- *Below the line:* Coal / Mining, Computers / Electronic Equipment / Semiconductors, Autos / Steel.
- *Near the line:* Wholesale / Retail, Transportation, Financials / Business Services.

Its reasoning was the low-beta anomaly. Defensive industries have low beta but still earn competitive returns, so they plot above the line, while cyclical industries have high beta without the returns to match. It added a sin-stock premium for tobacco and a glamour effect for technology. It stated that these were predictions from documented patterns, not calculations from the data.

(b) What the plot shows. Deviations are measured from the theoretical SML in Figure 1, as $\bar{R}_i - \bar{R}_M^e b_{iM}$.

Table 12: Largest deviations from the theoretical security market line, balanced sample, percent per month. A positive value means the industry earned more than its beta implies.

Above		Below		Near	
Smoke	+0.675	Other	−0.468	BldMt	−0.003
Guns	+0.461	RlEst	−0.429	Boxes	+0.004
Soda	+0.274	PerSv	−0.376	Telcm	+0.006

(c) How accurate were the predictions? The above-line predictions were accurate. All three industries plot above the line, and Tobacco is the largest positive deviation in the sample. The near-line predictions were also reasonable: Wholesale, Transportation and Financials all sit close to the line, though none is among the three closest.

The below-line predictions were mostly wrong. Coal and Mining plot slightly above the line rather than below it. Of the technology group, only Software is clearly below, while Chips and Electronic Equipment are above. Only Steel was placed correctly, and none of the three largest negative deviations was anticipated.

This asymmetry is explained by how the model reasoned. It used beta alone, predicting that low-beta industries plot above the line and high-beta industries plot below. Because the estimated line is so flat, the first half of this is almost automatic, since nearly any low-beta industry lands above the theoretical SML. That is why the above-line predictions worked. The below-line predictions needed more, because high beta is not sufficient: Chips and Electronic Equipment have high betas but earned enough to stay above the line. The industries that actually fall furthest below have betas near the sample average and unusually low returns, which reflects industry-specific weakness rather than a risk story the model could infer.

Its caveat about technology was partly right. It predicted technology would fall below the line but noted that recent data might reverse this. The reversal is what we observe for Chips and Electronic Equipment, though not for Software.

AI EXERCISE 3 — LITERATURE SYNTHESIS AND HISTORICAL CONTEXT

3(a)

Table 13: Fama-MacBeth estimates reported in Part I(b).

Sample	Parameter	Estimate	Standard error	t	p
July 1969–June 2026	γ_0	0.6119	0.2191	2.79	0.005
July 1969–June 2026	γ_M	0.0352	0.2725	0.13	0.897
July 1926–June 2026	γ_0	0.5160	0.1714	3.01	0.003
July 1926–June 2026	γ_M	0.2490	0.2315	1.08	0.282

Estimates and standard errors are in percent per month and the t -statistics test against zero. The mean market excess returns are 0.6116% and 0.6952% per month respectively. For the main sample the report gives $t = -2.12$ for testing the slope against the sample market premium.

My Part I(b) results weaken the favorable evidence of Fama and MacBeth (1973) and contradict the Sharpe–Lintner CAPM restrictions in my sample. For July 1969–June 2026, $\hat{\gamma}_0 = 0.6119\%$ per month ($SE = 0.2191$, $t = 2.79$, $p = 0.005$), significantly above the zero intercept predicted for excess returns. Meanwhile, $\hat{\gamma}_M = 0.0352\%$ per month ($SE = 0.2725$, $t = 0.13$, $p = 0.897$), providing no statistically significant evidence of a positive beta premium. The slope is also far below the CAPM benchmark of 0.6116%, with the reported benchmark test rejecting equality at the 5% level ($t = -2.12$). Thus, the estimated security market line is nearly flat, failing to replicate their positive beta–return relationship. The longer July 1926–June 2026 sample reinforces this pattern: the intercept remains significantly positive, while the beta premium remains insignificant. A positive excess-return intercept alone does not reject Black’s zero-beta CAPM, but the weak beta–return relationship undermines the earlier favorable interpretation.

3(b)

The original LLM response correctly identifies Banz's (1981) size effect: small firms earned higher average returns than their market betas could explain. It also accurately summarizes Fama and French (1992), emphasizing the explanatory power of size and book-to-market and the weak relation between beta and average returns once beta variation is separated from size. However, it omits Rosenberg, Reid, and Lanstein (1985), an important earlier study documenting the book-to-market effect that the assignment explicitly asks about. Although its discussion of Fama and French (1993) adds useful context, it does not fill this omission. Overall, the response captures the main empirical challenges to the CAPM but gives an incomplete account of the development of the value evidence.

3(c)

The original LLM response does not acknowledge data snooping: size and book-to-market may appear important partly because researchers identified and tested these patterns using the same historical data. This concern also applies to Part I(f), since our sample overlaps with the data used to discover these effects, so it is not a fully independent test. However, we use previously documented characteristics rather than search across many predictors ourselves. Our results provide only weak evidence: size is insignificant ($t = 0.86$), while book-to-market is marginally significant ($t = 1.95$). Lagging these characteristics addresses their timing but does not eliminate data snooping. Testing on a separate period after the original studies would provide stronger independent evidence.

3(d)

LLM response to record. International evidence shows that size and value effects occur outside the United States, although their strength varies across markets and samples.

- **Direct Fama–MacBeth evidence:** Hu, Chen, Shao, and Wang (2019) find a significant size-factor premium in Chinese stocks, while market and value-factor premia are insignificant. Thus, international Fama–MacBeth tests do not uniformly support both size and value. [1]
- **Broader international evidence:** Rouwenhorst (1999) finds that small stocks outperform large stocks and value stocks outperform growth stocks in emerging markets. Fama and French (1998) find value outperformance in twelve of thirteen major markets. These studies provide complementary evidence from portfolio comparisons and factor tests; their findings should not all be described as Fama–MacBeth regression results. [2], [3]

My interpretation is that international replication reduces the likelihood that these effects are merely artifacts of searching U.S. data. It is consistent with compensation for omitted risks, but does not establish that explanation: similar investor biases and limits to arbitrage could also produce recurring patterns across countries.

Evaluation for submission. The international evidence modestly strengthens the plausibility of a risk-based interpretation by showing that size and value patterns are not confined to U.S. data. However, it strengthens the evidence that these patterns exist more clearly than it distinguishes risk from mispricing, since investor biases and limits to arbitrage may also operate internationally. The LLM appropriately acknowledges this distinction and the variation across markets. In our Part I(f), size is insignificant ($t = 0.86$) and book-to-market is only marginally significant ($t = 1.95$), so international findings cannot turn our weak evidence into proof of priced risks. They provide supporting context for the positive book-to-market coefficient, but establishing a risk explanation would require evidence linking returns to exposure to economically meaningful risks.

Sources cited in 3(d). [1] Hu, Chen, Shao, and Wang (2019). “Fama–French in China: Size and Value Factors in Chinese Stock Returns.” doi:10.1111/irfi.12177
 [2] Rouwenhorst (1999). “Local Return Factors and Turnover in Emerging Stock Markets.” doi:10.1111/0022-1082.00151
 [3] Fama and French (1998). “Value versus Growth: The International Evidence.” doi:10.1111/0022-1082.00080

CAPSTONE EXERCISE: DESIGN YOUR OWN CROSS-SECTIONAL TEST

Step 1 — Choose a Dataset

Of the datasets offered, this capstone uses **U.S. sector ETF returns**.

Research question. Does market beta explain cross-sectional differences in average returns across U.S. sector ETFs?

Step 2 — Formulate a Hypothesis

The dependent variable is the average monthly return of each U.S. sector ETF, and the main independent variable is its market beta relative to SPY.

I expect sector ETFs with higher market betas to have higher average returns. Under the CAPM, the relationship should be positive, with the coefficient on beta approximately reflecting the market risk premium. Economically, investors should require higher expected returns for bearing greater systematic market risk.

I would view the results as supportive of the CAPM if the beta coefficient is positive and statistically significant, with an intercept reasonably close to the risk-free rate. If the beta coefficient is insignificant, zero, or negative, or if the intercept differs substantially from the risk-free rate, I would conclude that the CAPM does not explain the cross-section of sector ETF returns well.

Specification clarification (added after analysis). The implemented regression uses excess returns rather than raw returns. Therefore, the Sharpe-Lintner CAPM predicts an intercept of zero, rather than the risk-free rate. The original hypothesis above is retained as written before examining the data.

Step 3 — Use AI to Help Implement the Test

I use monthly returns for nine U.S. sector ETFs (XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, and XLY), with SPY as the market proxy and the risk-free rate from the Kenneth French Data Library.

The sample covers February 2000 through December 2025, with 311 monthly observations.

I first calculate excess returns by subtracting the monthly risk-free rate from each ETF’s return and SPY’s return. I then estimate each ETF’s market beta using a time-series regression of ETF excess returns on SPY excess returns.

Finally, I run a cross-sectional OLS regression of average ETF excess returns on their estimated market betas. This regression provides the coefficient estimates. I use Fama-MacBeth (FMB) standard errors as the main basis for inference because OLS standard errors treat the nine ETF residuals as independent, although sector returns share common monthly shocks.

Each month, I regress the nine ETF excess returns on the same fixed full-sample betas, then take the time-series average of the 311 monthly intercept and slope estimates. The FMB standard error is the time-series standard deviation of the monthly estimates divided by $\sqrt{T}$. Because the betas are the same every month, the FMB coefficient averages equal the OLS coefficients exactly; only the standard errors differ.

AI workflow log.

Step 3(a): Data Loading and Cleaning

Prompt (Close Paraphrase). I asked the LLM to generate Python code to download monthly adjusted-price data for nine U.S. sector ETFs (XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY) and SPY from January 2000 through December 2025, calculate monthly returns, align the series, and check for missing observations.

LLM Output and My Correction. The code successfully downloaded the data and produced 311 monthly return observations from February 2000 through December 2025.

However, it checked for missing values only after dropping missing observations, making the diagnostic uninformative. I revised the code to report missing values before and after cleaning and explicitly used `fill_method=None` when calculating returns. The final dataset contained no missing prices, so no observations were removed.

Step 3(b): Cross-Sectional Regression

Initial OLS Regression

Prompt (Close Paraphrase). Please generate Python code to estimate the market beta of each of the nine U.S. sector ETFs using SPY as the market proxy. Then run a cross-sectional OLS regression of each ETF's average monthly return on its estimated beta. Report the coefficients, standard errors, *t*-statistics, *p*-values, and R-squared.

LLM Output and My Correction. The LLM generated code to estimate full-sample betas and regress average ETF returns on those betas. The initial results were an intercept of 0.0061, a beta coefficient of 0.0020 ($p = 0.0721$), and $R^2 = 0.3903$.

However, the code used raw rather than excess returns. I revised it to subtract the monthly risk-free rate from both ETF and SPY returns, using data from the Kenneth French Data Library. This matches the Sharpe-Lintner CAPM specification, which predicts a zero intercept for excess returns.

Fama-MacBeth Robustness Check

A later AI code review recommended Fama-MacBeth (FMB) standard errors because the original cross-sectional OLS inference did not adequately account for common shocks across sector returns.

Prompt (Close Paraphrase). Keep my original cross-sectional OLS regression as the main methodology. Add Fama-MacBeth standard errors only as a robustness check, using the same fixed full-sample betas and monthly ETF excess returns. Report the OLS and FMB results side by side, including coefficients, standard errors, *t*-statistics and *p*-values, and compare the estimated beta premium with the average SPY excess return. Do not add Shanken, Newey-West or other extensions.

LLM Output and Verification. The LLM generated monthly cross-sectional regressions using fixed betas, then calculated FMB standard errors from the time-series variation in the 311 monthly coefficient estimates.

I reran the script and verified that FMB and OLS produced identical coefficient estimates but different standard errors, *t*-statistics, and *p*-values. With FMB standard errors, neither the intercept ($p = 0.1099$) nor the beta coefficient ($p = 0.5823$) is significant. I therefore use the FMB standard errors as the main basis for inference.

Additional Correction: Market-Premium Gap Test. The initial comparison divided the gap by the beta premium's standard error alone, ignoring uncertainty in the SPY average and its covariance with the estimated premium.

I replaced it with a test based on the monthly difference series between the estimated beta premium and SPY excess returns. The gap is -0.0040 , with $SE = 0.0027$, $t = -1.47$ and $p = 0.1416$. The difference is not statistically significant.

Step 3(c): Interpretation Assistance

Initial Raw-Return Results

Prompt. I ran a cross-sectional CAPM test on nine U.S. sector ETFs using monthly data from February 2000 to December 2025. The cross-sectional regression of average monthly returns on estimated market beta produced an intercept of 0.0061 ($t = 6.47$, $p = 0.0003$), a beta coefficient of 0.0020 ($t = 2.12$, $p = 0.0721$), and an R-squared of 0.3903 . Please interpret these results. Do they support my hypothesis that higher-beta sector ETFs should have higher average returns, and how strong is the evidence for the CAPM?

Summary of the LLM's Answer. The LLM correctly interpreted the positive beta coefficient as consistent with my hypothesis. It noted that the coefficient was significant at the 10% level but not at 5%, and that beta explained approximately 39% of the observed cross-sectional variation. It concluded that the evidence for the CAPM was limited.

My Evaluation. The LLM correctly interpreted the coefficient's sign, magnitude, and significance. However, it overlooked the intercept and failed to compare the estimated slope with the market risk premium, both of which are central CAPM restrictions.

What it got wrong: it relied on OLS p -values without accounting for common shocks across sector returns and did not emphasize the small cross-section or estimated-beta uncertainty. Some omissions reflected my original prompt, which did not provide the risk-free rate or market premium.

Re-assessment After Code Corrections

Switching to excess returns reduced the intercept from 0.0061 to 0.0045 , while the beta coefficient remained approximately 0.0020 . The subsequent FMB check showed that neither coefficient was significant, and the beta premium was not significantly different from the SPY premium ($p = 0.1416$).

These additional checks came from a later AI code review, not from the original LLM interpretation. They changed my initial interpretation of the significant OLS intercept as evidence against the CAPM.

Final Excess-Return Results

Prompt. I ran a cross-sectional CAPM test using nine U.S. sector ETFs, with SPY as the market proxy. My sample contains 311 monthly observations from February 2000 through December 2025.

I estimated each ETF's beta by regressing its excess returns on SPY excess returns. I then regressed average ETF excess returns on estimated beta. I also calculated Fama-MacBeth standard errors using monthly cross-sectional regressions with fixed betas.

Here are my final results (all coefficients and standard errors are in decimal units per month):

	Intercept	Beta
Coefficient	0.0045	0.0020
OLS SE	0.0009	0.0010
OLS t -stat	4.8158	2.1220
OLS p -value	0.0019	0.0715
FMB SE	0.0028	0.0037
FMB t -stat	1.6032	0.5506
FMB p -value	0.1099	0.5823

Cross-sectional R-squared: 0.3915

Number of ETFs: 9

Average monthly SPY excess return: 0.0060

The estimated beta premium minus the average SPY excess return is -0.0040 . A test based on the monthly difference series gives $t = -1.47$ and $p = 0.1416$.

Please interpret these results. Do they support my hypothesis that higher-beta sector ETFs earn higher average excess returns? What do they imply about the CAPM, and how do the conclusions differ when using OLS versus Fama-MacBeth standard errors?

Summary of the LLM's Answer. The LLM correctly noted that the positive beta coefficient is consistent with my hypothesis but insignificant at the 5% level under both OLS and FMB inference.

It explained that the positive intercept and relatively low beta premium suggest a flatter SML than the CAPM predicts. Although the OLS intercept is significant, neither coefficient is significant using FMB standard errors. The gap between the beta premium and the SPY premium is also insignificant.

It concluded that the point estimates suggest a flatter SML but do not provide sufficient evidence to reject the individually tested CAPM restrictions. It also noted the small cross-section, estimated-beta uncertainty, potential serial correlation, and absence of a formal joint test.

My Evaluation. The LLM correctly distinguished OLS from FMB inference and compared the estimated beta premium with the market premium rather than simply testing whether beta is significant.

However, it did not fully explain the implications of the test's low precision. The approximate 95% FMB confidence interval for the beta premium is -0.53% to 0.93% per month, including both zero and the SPY premium of 0.60% .

It also mentioned estimated-beta uncertainty without explaining that measurement error can flatten the estimated SML. Finally, it overlooked Roll's critique: SPY is only a proxy for the true market portfolio. I found no major numerical error in its interpretation, but these limitations deserved more attention.

Code. The commented, runnable script used to load the data, run the regressions and generate the results table is reproduced in the appendix at the end of this report (Capstone Deliverable 2).

Step 4 — Report Your Results

Table 14: Cross-sectional CAPM test on nine sector ETFs.

Variable	Coefficient	OLS Std. Error	OLS t -stat	OLS p -value	FMB Std. Error	FMB t -stat	FMB p -value
Intercept	0.0045	0.0009	4.8158	0.0019	0.0028	1.6032	0.1099
Beta	0.0020	0.0010	2.1220	0.0715	0.0037	0.5506	0.5823

- Number of sector ETFs: 9
- Sample period: February 2000–December 2025
- Monthly return observations (T): 311
- Cross-sectional R^2 : 0.3915
- Average SPY excess return: 0.0060
- OLS p -values use $N - 2 = 7$ degrees of freedom; FMB p -values use $T - 1 = 310$ degrees of freedom.

All coefficients and standard errors are reported in decimal units per month. For comparison with Part I, which reports returns in percent per month, the estimates should be multiplied by 100.

Table 15: Beta premium vs. market premium.

	Value
Estimated beta premium (γ_M)	0.0020
Average SPY excess return	0.0060
Difference ($\gamma_M - \text{SPY}$)	-0.0040
Std. error of difference (monthly series)	0.0027
t -stat of difference	-1.47
p -value of difference	0.1416

The test of the difference uses the monthly series $\gamma_{Mt} - \text{SPY}_t$, whose time-series standard error accounts for the sampling uncertainty of both averages and their covariance. There is no equivalent monthly series for the single cross-sectional OLS regression, so under OLS the comparison is reported descriptively only.

(a) Do the results support the hypothesis? The estimated beta premium is positive at 0.20% per month, consistent with my hypothesis, but insignificant using Fama-MacBeth standard errors ($p = 0.5823$). It is below the average SPY excess return of 0.60%, while the estimated intercept is 0.45% rather than zero, suggesting a flatter SML than the CAPM predicts. Although OLS finds a significant intercept ($p = 0.0019$), the FMB intercept test ($p = 0.1099$) and the paired market-premium gap test ($p = 0.1416$) are insignificant. By my Step 2 criterion, the hypothesis is not supported because the beta premium is insignificant. However, an insignificant slope is not by itself a rejection of the CAPM, and the FMB tests do not reject its intercept or slope restrictions.

(b) Comparison with Parts I and II. My estimated beta premium of 0.20% per month compares with 0.035% for the 49 industry portfolios and -0.576% for the 25 size/BE-ME portfolios. All three are insignificant using FMB standard errors and below their respective market premiums. However, Parts I and II find significantly positive intercepts and significant differences between their estimated beta premiums and market premiums, while my ETF test does not. Part II also finds a significant book-to-market premium after controlling for beta. The slope is positive for Part I and my Capstone and negative for Part II. So the ETF cross-section looks similar to US equities (a flat SML), but my test gives much less precise evidence against the CAPM.

(c) Main limitation. The main limitation is the small cross-section of nine sector ETFs. Their relatively similar market exposures limit variation in beta, reducing the test's statistical power. FMB standard errors are substantially larger than OLS standard errors, and the estimated beta premium cannot be statistically distinguished from the market premium. In addition, using estimated rather than true betas can bias the slope toward zero, while SPY is only a proxy for the true market portfolio.

Step 5 — AI Workflow Reflection

Using an LLM helped me move quickly from my research design to working Python code, allowing me to spend more time checking the results. I corrected the missing-value diagnostics and changed the regression from raw to excess returns to match the Sharpe-Lintner CAPM. A later AI review prompted me to add Fama-MacBeth standard errors, which changed my interpretation of the statistical evidence. These experiences showed me that AI can help both generate code and question

results, but I still need to verify its methods and conclusions independently. Next time, I would specify the CAPM form, excess returns, data checks, and inference method in my initial prompts.

APPENDIX: CAPSTONE ANALYSIS SCRIPT

Capstone Deliverable 2 asks for the commented, runnable script used to load the data, run the regressions and generate the results table. The submission accepts a single PDF, so the script is reproduced here in full. The code for Parts I and II is held in the project repository rather than printed.

```

1  import yfinance as yf
2  import numpy as np
3  import pandas as pd
4  import statsmodels.api as sm
5  from scipy import stats
6  import io
7  import zipfile
8  import requests
9
10 # U.S. sector ETFs and market proxy
11 sector_etfs = [
12     "XLB", "XLE", "XLF", "XLI", "XLK",
13     "XLP", "XLU", "XLV", "XLY"
14 ]
15 market = "SPY"
16
17 tickers = sector_etfs + [market]
18
19 # Download monthly adjusted price data
20 prices_raw = yf.download(
21     tickers,
22     start="2000-01-01",
23     end="2026-01-01",
24     interval="1mo",
25     auto_adjust=True,
26     progress=False
27 )["Close"]
28
29 # Check missing values before cleaning
30 print("Missing prices before cleaning:")
31 print(prices_raw.isna().sum())
32
33 print("\nNumber of price observations before cleaning:", len(prices_raw))
34
35 # Keep a common sample where all ETFs have price data
36 prices = prices_raw.dropna()
37
38 print("Number of price observations after cleaning:", len(prices))
39
40 # Calculate monthly returns
41 returns = prices.pct_change(fill_method=None).dropna()
42
43 # Basic checks
44 print("\nSample start:", returns.index.min())
45 print("Sample end:", returns.index.max())
46 print("Number of monthly return observations:", len(returns))
47
48 print("\nMissing returns after cleaning:")
49 print(returns.isna().sum())
50
51 print("\nFirst five rows:")
52 print(returns.head())
53
54 # -----
55 # Step 3(b): CAPM test using excess returns
56 # -----

```

```

57
58 # Download monthly risk-free rate from the
59 # Kenneth French Data Library
60 ff_url = (
61     "https://mba.tuck.dartmouth.edu/pages/faculty/"
62     "ken.french/ftp/F-F_Research_Data_Factors_CSV.zip"
63 )
64
65 response = requests.get(ff_url, timeout=30)
66 response.raise_for_status()
67
68 with zipfile.ZipFile(io.BytesIO(response.content)) as z:
69     file_name = z.namelist()[0]
70     with z.open(file_name) as f:
71         lines = f.read().decode("utf-8").splitlines()
72
73 # Locate the monthly factor table
74 header_idx = next(
75     i for i, line in enumerate(lines)
76     if "Mkt-RF" in line and "RF" in line
77 )
78
79 monthly_rows = []
80
81 for line in lines[header_idx + 1:]:
82     first_value = line.split(",")[0].strip()
83
84     if len(first_value) == 6 and first_value.isdigit():
85         monthly_rows.append(line)
86     elif monthly_rows:
87         break
88
89 ff = pd.read_csv(
90     io.StringIO(
91         "\n".join([lines[header_idx]] + monthly_rows)
92     )
93 )
94
95 ff.rename(columns={ff.columns[0]: "Date"}, inplace=True)
96
97 ff["Date"] = pd.PeriodIndex(
98     ff["Date"].astype(str),
99     freq="M"
100 )
101
102 ff = ff.set_index("Date")
103
104 # French data are reported in percent, so convert RF to decimals
105 rf = ff["RF"] / 100
106
107
108 # -----
109 # Align ETF returns and risk-free rate
110 # -----
111
112 returns_monthly = returns.copy()
113 returns_monthly.index = returns_monthly.index.to_period("M")
114
115 common_dates = returns_monthly.index.intersection(rf.index)
116
117 returns_aligned = returns_monthly.loc[common_dates]
118 rf_aligned = rf.loc[common_dates]
119
120 # Compute excess returns
121 excess_returns = returns_aligned.sub(rf_aligned, axis=0)
122
123 market_excess = excess_returns[market]
124

```

```

125 print("\nCAPM sample after aligning risk-free rate:")
126 print("Start:", common_dates.min())
127 print("End:", common_dates.max())
128 print("Number of months:", len(common_dates))
129 print(
130     "Average monthly risk-free rate:",
131     round(rf_aligned.mean(), 4)
132 )
133 print(
134     "Average monthly SPY excess return:",
135     round(market_excess.mean(), 4)
136 )
137
138
139 # -----
140 # First pass: estimate each sector ETF's market beta
141 # -----
142
143 beta_results = []
144
145 for ticker in sector_etfs:
146     y = excess_returns[ticker]
147     X = sm.add_constant(market_excess)
148
149     model = sm.OLS(y, X).fit()
150
151     beta_results.append({
152         "Ticker": ticker,
153         "Average Excess Return": y.mean(),
154         "Beta": model.params[market]
155     })
156
157 beta_table = pd.DataFrame(beta_results)
158
159 print("\nSector ETF beta estimates:")
160 print(beta_table.round(4))
161
162
163 # -----
164 # Second pass: cross-sectional regression
165 #
166 # Average Excess Returni =
167 #  $\gamma_0 + \gamma_M * \text{Beta}_i + \text{error}_i$ 
168 # -----
169
170 y_cs = beta_table["Average Excess Return"]
171 X_cs = sm.add_constant(beta_table["Beta"])
172
173 cs_model = sm.OLS(y_cs, X_cs).fit()
174
175 results_table = pd.DataFrame({
176     "Coefficient": cs_model.params,
177     "Std. Error": cs_model.bse,
178     "t-stat": cs_model.tvalues,
179     "p-value": cs_model.pvalues
180 })
181
182 print("\nCross-sectional CAPM regression:")
183 print(results_table.round(4))
184
185 print(
186     "\nCross-sectional R-squared:",
187     round(cs_model.rsquared, 4)
188 )
189 print(
190     "Number of sector ETFs:",
191     len(beta_table)
192 )

```

```

193
194
195 # -----
196 # Fama-MacBeth standard errors (main basis for inference)
197 #
198 # Each month  $t$ , regress the nine ETF excess returns on
199 # the same fixed full-sample betas:
200 #  $R_{it} = \gamma_{0t} + \gamma_{Mt} * \beta_i + \text{error}_{it}$ 
201 #
202 # The FMB estimate is the time-series average of the
203 # monthly gammas; its standard error is the time-series
204 # standard deviation of the gammas divided by  $\sqrt{T}$ .
205 # -----
206
207 X_fmb = sm.add_constant(beta_table.set_index("Ticker")["Beta"])
208
209 monthly_gammas = []
210
211 for date in excess_returns.index:
212     y_t = excess_returns.loc[date, sector_etfs]
213     monthly_gammas.append(sm.OLS(y_t, X_fmb).fit().params)
214
215 monthly_gammas = pd.DataFrame(monthly_gammas, index=excess_returns.index)
216
217 T = len(monthly_gammas)
218 fmb_coef = monthly_gammas.mean()
219 fmb_se = monthly_gammas.std(ddof=1) / np.sqrt(T)
220 fmb_t = fmb_coef / fmb_se
221 fmb_p = 2 * stats.t.sf(np.abs(fmb_t), df=T - 1)
222
223 # The FMB averages should equal the OLS coefficients,
224 # because the betas are the same every month
225 assert np.allclose(fmb_coef, cs_model.params)
226
227 comparison_table = pd.DataFrame({
228     "OLS Coef.": cs_model.params,
229     "OLS Std. Error": cs_model.bse,
230     "OLS t-stat": cs_model.tvalues,
231     "OLS p-value": cs_model.pvalues,
232     "FMB Coef.": fmb_coef,
233     "FMB Std. Error": fmb_se,
234     "FMB t-stat": fmb_t,
235     "FMB p-value": fmb_p
236 })
237
238 print("\nOLS vs. Fama-MacBeth (main basis for inference):")
239 print(comparison_table.round(4).T)
240 print("Number of monthly cross-sections (T):", T)
241
242
243 # -----
244 # Compare the estimated beta premium with the
245 # average SPY excess return (the CAPM prediction)
246 # -----
247
248 beta_premium = cs_model.params["Beta"]
249 spy_premium = market_excess.mean()
250 premium_gap = beta_premium - spy_premium
251
252 print("\nBeta premium vs. average SPY excess return:")
253 print("Estimated beta premium (gamma_M):", round(beta_premium, 4))
254 print("Average SPY excess return:", round(spy_premium, 4))
255 print("Difference (gamma_M - SPY):", round(premium_gap, 4))
256
257 # Test the difference using the monthly series  $\gamma_{Mt} - \text{SPY}_t$ .
258 # Its time-series standard error accounts for the sampling
259 # uncertainty of both averages and their covariance.
260 monthly_gap = monthly_gammas["Beta"] - market_excess

```

```
261 gap_se = monthly_gap.std(ddof=1) / np.sqrt(T)
262 gap_t = monthly_gap.mean() / gap_se
263 gap_p = 2 * stats.t.sf(abs(gap_t), df=T - 1)
264
265 assert np.isclose(monthly_gap.mean(), premium_gap)
266
267 print("Std. error of difference (monthly series):", round(gap_se, 4))
268 print("t-stat of difference:", round(gap_t, 4))
269 print("p-value of difference:", round(gap_p, 4))
```